

Lctr 3: LTI systems, convolution, and LCCDEs

Learning outcomes:

- *LO5. Calculate the convolution sum.*
- *LO6. Compute output from input for systems described by LCCDE.*
- *LO7. List LCCDE system properties.*

LO5. Calculate the convolution sum.

- Convolution: Convolution sum is the most general description of the input-output behavior of linear, time-invariant systems.



- Impulse response $h[n]$: Response of the system to the unit impulse $\delta[n]$; i.e.,
$$x[n] = \delta[n] \Rightarrow y[n] = h[n].$$

- From LTI to convolution with an example:

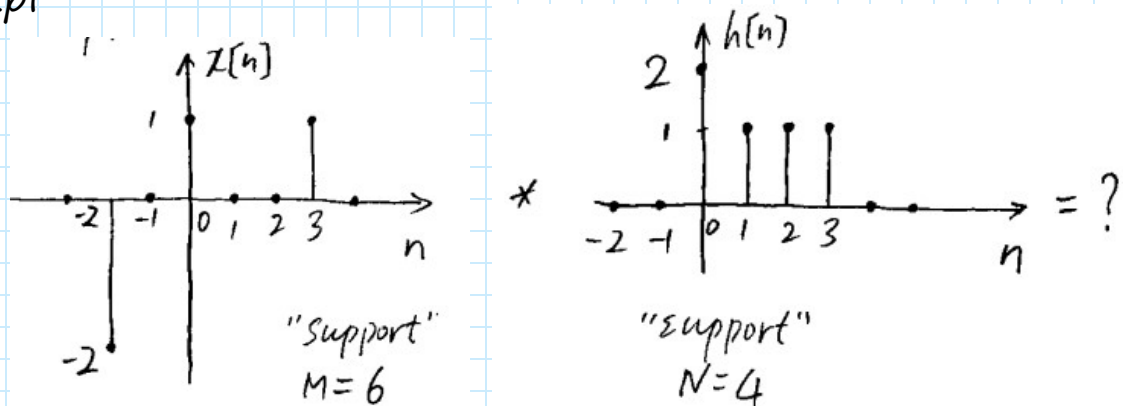
- Convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n]$$

↑
"convolved with"

- Calculation of convolution sum when input has only a few non-zero terms

Expl



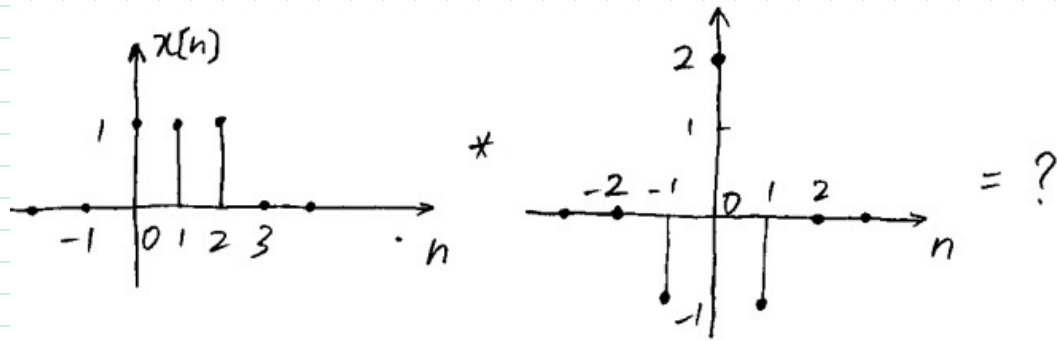
Soln:

$$x[n] = -2\delta[n+2] + \delta[n] + \delta[n-3]$$

$$\Rightarrow y[n] = -2h[n+2] + h[n] + h[n-3]$$

n	-4	-3	-2	-1	0	1	2	3	4	5	6	7
$h[n]$	0	0	0	0	2	1	1	1	0	0	0	0
$2h[n+2]$	0	0	-4	-2	-2	-2	0	0	0	0	0	0
$h[n]$	0	0	0	0	2	1	1	1	0	0	0	0
$h[n-3]$												
$y[n]$												

Exercise:



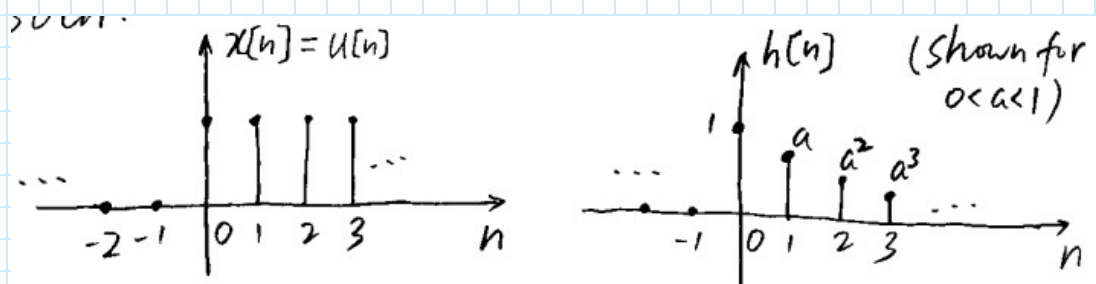
- Calculation of convolution sum for longer or infinite-duration signals:

Flip, Slide, and Sum

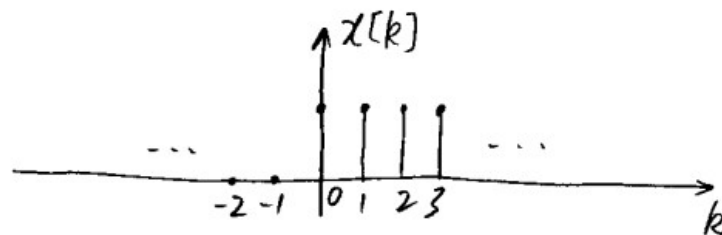
$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

Expl: Determine the step response of the system with impulse response $a^n u[n]$.

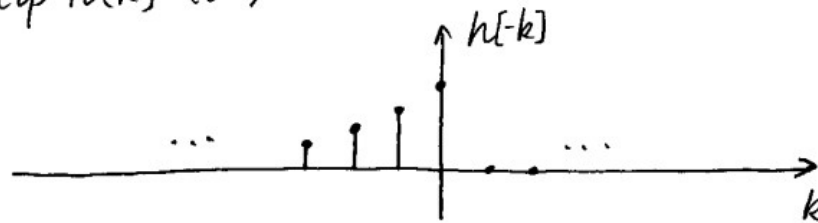
Soln:



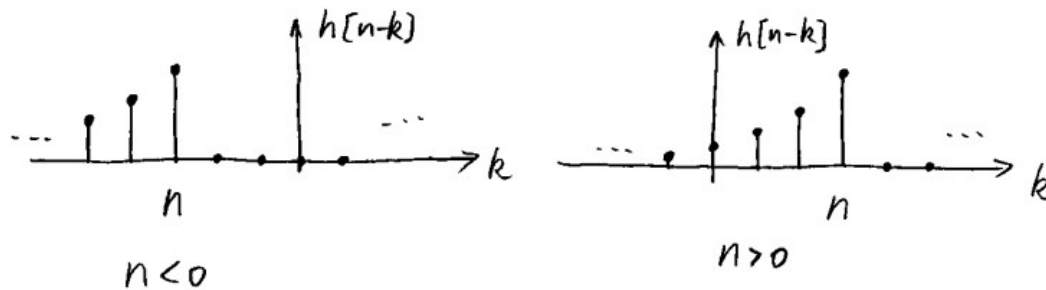
- Draw $x[k]$



◦ Flip $h[k]$ to get $h[-k]$



◦ Slide $h[-k]$ to obtain $h[n-k]$ (n is a parameter now)



($h[n-k]$ is $h[-k]$ shifted to right by n)

Remarks: Conceptually, we need to do this for every value of $n \in \mathbb{I}$; but usually, we just articulate the different overlap regions and then set up simplified summation.

□ For $n < 0$, the non-zero parts of $x[k]$ and $h[n-k]$ do not overlap $\Rightarrow$

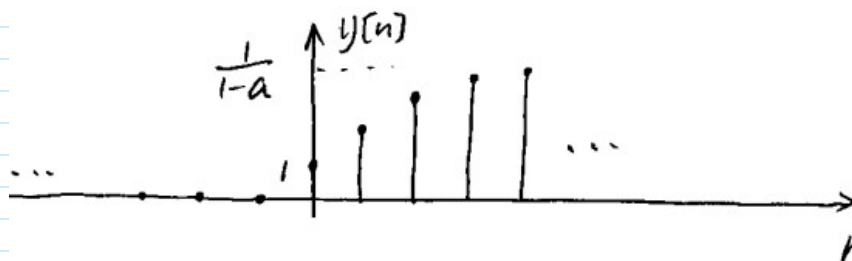
$$y[n] = 0, \quad n < 0$$

□ For $n \geq 0$, $x[k]$ is zero for $k < 0$ and $h[n-k]$ is zero for $k > n \Rightarrow$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k] = \sum_{k=0}^n 1 \cdot a^{n-k}$$

$$= a^n \sum_{k=0}^n (a^{-1})^k = a^n \cdot \frac{1 - (a^{-1})^{n+1}}{1 - a^{-1}}$$

$$= \frac{1 - a^{n+1}}{1 - a}$$



□

- Convolution properties:

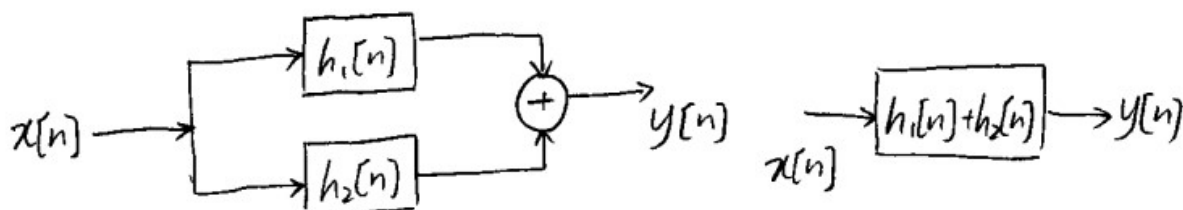
- Commutative

$$\begin{aligned} x[n] * h[n] &= \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k] && \text{(Let } n-k=m \\ &&& \text{then } k=n-m) \\ &= \sum_{m=-\infty}^{\infty} x[n-m] \cdot h[m] \\ &= \sum_{k=-\infty}^{\infty} h[k] x[n-k] && \text{(Let } m=k) \\ &= h[n] * x[n]. \end{aligned}$$

Remarks: we can choose which signal to flip and slide.

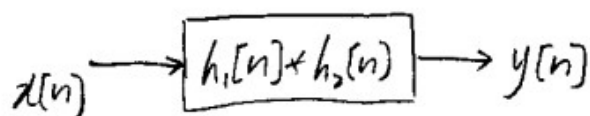
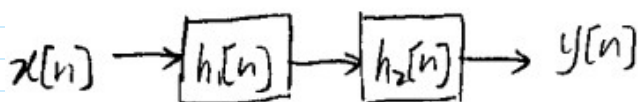
- Distributive over addition

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n].$$



- Associative

$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$$



Other results:

- An LTI system is causal if $h[n] \equiv 0$ for $\forall n < 0$
- An LTI system is BIBO stable when

$$\sum_n |h[n]| < \infty$$

($h[n]$ is said to be absolutely summable)